AIR UNIVERSITY

DEPARTMENT OF ELECTRICAL AND COMPUTER ENGINEERING

EXPERIMENT NO 7

Lab Title Convolution

Student Name: Ali Abbas Khan

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Objective:

LAB ASSESSMENT:

Attributes	Excellent (5)	Good (4)	Average (3)	Satisfactory (2)	Unsatisfactory (1)
Ability to Conduct Experiment					
Ability to assimilate the results					
Effective use of lab equipment and follows the lab safety rules					

Total Marks: _____ Obtained Marks: _____

LAB REPORT ASSESSMENT:

Attributes	Excellent (5)	Good (4)	Average (3)	Satisfactory (2)	Unsatisfactory (1)
Data presentation					
Experimental results					
Conclusion					

Total Marks: _____ Obtained Marks: _____

Date: _____ Signature: _____



Department of Electrical and Computer Engineering

Signal and Systems Lab

Lab Report: 7

Title: Convolution

Name: Ali Abbas Khan

Roll Number: 230661

Class: BCE-5B

Submitted To: Engr. Maryam
Date**:** November 14, 2025

Lab Tasks:

Q1 Suppose that a linear time-invariant system is described by impulse response $h[n] = 4^n * u[n-1]$. Compute response of the system to the input signal $x[n] = u[n-10] - u[n+5]$ for $n = -15:15$. Hint: ($u[n]$ is the unit step function. Use the 'conv' function for computing the convolution of the given signals and use subplot() command to plot $x[n]$, $h[n]$ and $y[n]$.

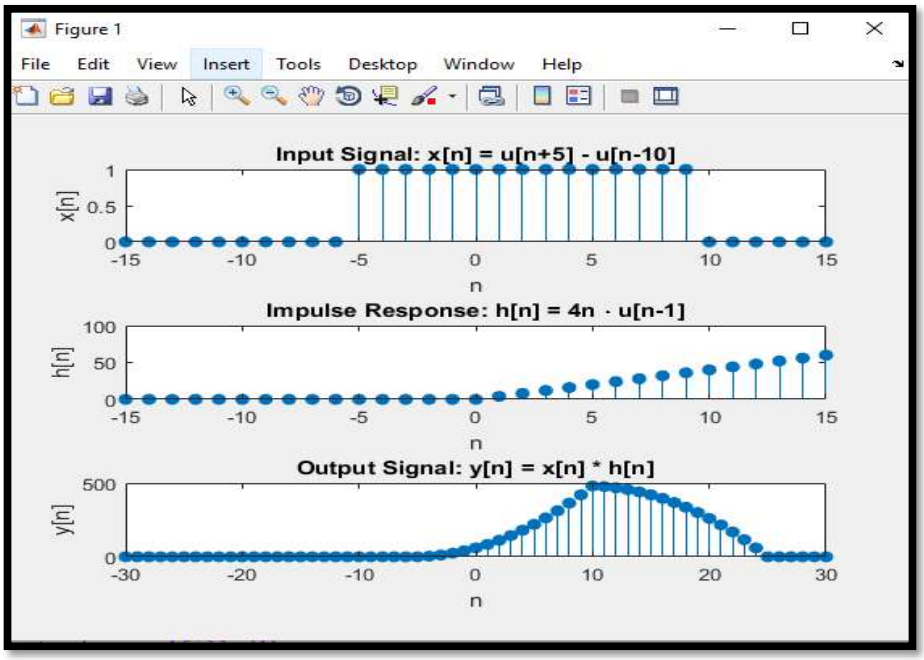
Code:

Editor - C:\Users\Admin\OneDrive\Desktop\SNS Lab\Matlab Files\L7\T1.m

T1.m | T2.m | T3.m | T4.m

```
n=-15:15;
u1=(n+5>=0);
u2=(n-10>=0);
xn=u1-u2;
hn=4*n.*(n >= 1);
yn=conv(xn,hn);
ny=(min(n)+min(n)):(max(n)+max(n));
figure;
subplot(3,1,1);
stem(n,xn,'filled');
xlabel('n'); ylabel('x[n]');
title('Input Signal: x[n] = u[n+5] - u[n-10]');
subplot(3,1,2);
stem(n,hn,'filled');
xlabel('n'); ylabel('h[n]');
title('Impulse Response: h[n] = 4n \cdot u[n-1]');
subplot(3,1,3);
stem(ny,yn,'filled');
xlabel('n'); ylabel('y[n]');
title('Output Signal: y[n] = x[n] * h[n]');
```

Output:



Q2 Using MATLAB:

Compute the convolution y[n]=x[n]*h[n].

a. Use the subplot command to plot the following:
b. The input signal x[n]
c. The impulse response h[n]
d. The output signal y[n].

$$x[n] = \begin{cases} 1, & 0 \le n \le 5 \\ 0, & \text{otherwise} \end{cases}$$

$$h[n] = \begin{cases} e^{-n}, & -10 \le n \le 10 \\ 0, & \text{otherwise} \end{cases}$$

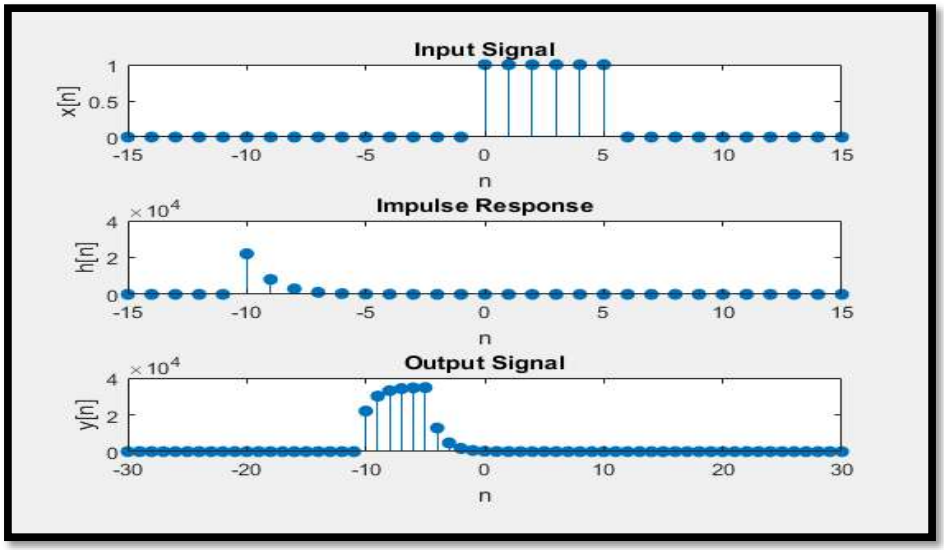
Code:

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T1.m | T2.m | T3.m | T4.m

```
n=-15:15;
xn=(n>=0)&(n<=5);
hn_base=exp(-n);
mask=(n>=-10)&(n<=10);
hn=hn_base.*mask;
yn=conv(xn,hn);
start_n=min(n)+min(n);
end_n=max(n)+max(n);
ny=start_n:end_n;
figure;
subplot(3,1,1);
stem(n,xn,'filled');
xlabel('n'); ylabel('x[n]');
title('Input Signal');
subplot(3,1,2);
stem(n,hn,'filled');
xlabel('n'); ylabel('h[n]');
title('Impulse Response');
subplot(3,1,3);
stem(ny,yn,'filled');
xlabel('n'); ylabel('y[n]');
title('Output Signal');
```

Output:



Q3 Consider the following input signal and the impulse response of the LTI system:

$$x[n] = e^{-2n}u[n]$$

$$h[n] = u[n-10]$$

Find the output y[n] when n=-15:15. Use subplot command to plot $x[n]$,$h[n]$ and $y[n]$.

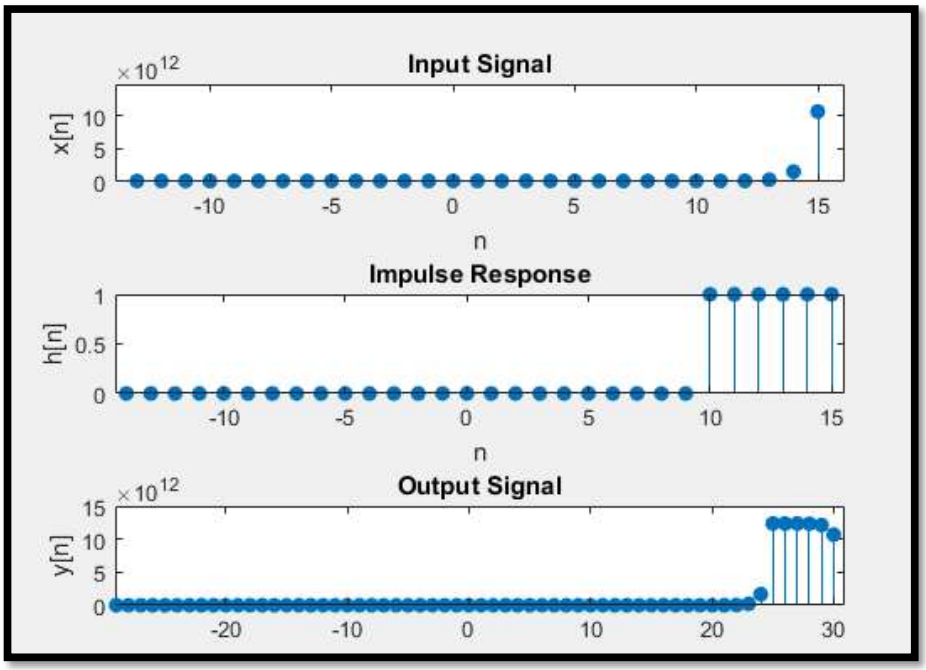
Code:

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T1.m | T2.m | T3.m | T4.m

```
n=-15:15;
xn_base=exp(2*n);
mask=(n>=0);
xn=xn_base.*mask;
hn=(n>=10);
yn=conv(xn,hn);
ny=(min(n)+min(n)):(max(n)+max(n));
figure;
subplot(3,1,1);
stem(n,xn,'filled');
xlabel('n'); ylabel('x[n]');
title('Input Signal');
subplot(3,1,2);
stem(n,hn,'filled');
xlabel('n'); ylabel('h[n]');
title('Impulse Response');
subplot(3,1,3);
stem(ny,yn,'filled');
xlabel('n'); ylabel('y[n]');
title('Output Signal');
```

Output:



Q4 Use conv() to compute the convolution and plot x[n], h[n], and y[n].

$$x[n] = u[n] - u[n-4]$$

$$h[n] = (0.5)^n u[n]$$

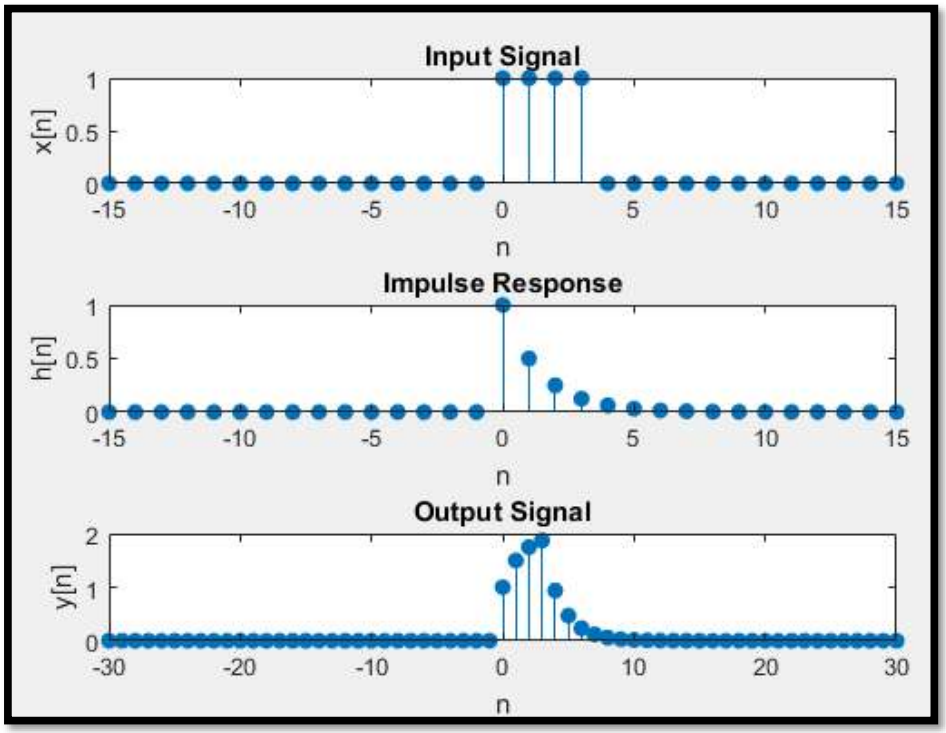
Code:

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T1.m | T2.m | T3.m | T4.m

```
n=-15:15;
xn=(n>=0)-(n>=4);
hn_base=0.5.^n;
mask=(n>=0);
hn=hn_base.*mask;
yn=conv(xn,hn);
ny=(min(n)+min(n)):(max(n)+max(n));
figure;
subplot(3,1,1);
stem(n,xn,'filled');
xlabel('n'); ylabel('x[n]');
title('Input Signal');
subplot(3,1,2);
stem(n,hn,'filled');
xlabel('n'); ylabel('h[n]');
title('Impulse Response');
subplot(3,1,3);
stem(ny,yn,'filled');
xlabel('n'); ylabel('y[n]');
title('Output Signal');
```

Output:



Conclusion:

Convolution in MATLAB is handled by the simple conv function, which is the tool for simulating how an electrical circuit, a filter, or any system changes an input signal. You provide the function with two main ingredients: the input signal and the system's impulse response. The output is a new, single signal that represents the system's reaction. To properly plot this result, you must carefully define the time axis for the final signal. This axis starts at the sum of the input signal's start time and the impulse response's start time, and it ends at the sum of their end times, ensuring the plot captures the entire spread-out and combined effect of the two initial signals.